

Playing Tougher for One's Size: Positional Physicality in the NHL

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**Statistics and Data Science, Computer Science
Data, Information, and Society Pathway
Funded Sophomore Research Program**

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What does CSAx mean?

**Height and weight describe a player's frame.
How they play can tell a different story.**

We use recorded NHL actions to ask: does this skater play with more physical presence than their size suggests?

DIRECT CONTACT

Hits given or taken, blocks, fights, contact penalties

PUCK + SPACE

Net-front plays and forechecking; defensive recoveries and entry denials

Higher CSAx

More physical presence relative to listed size

A lower score describes less measured physicality for one's frame—not skill, courage, or overall value.

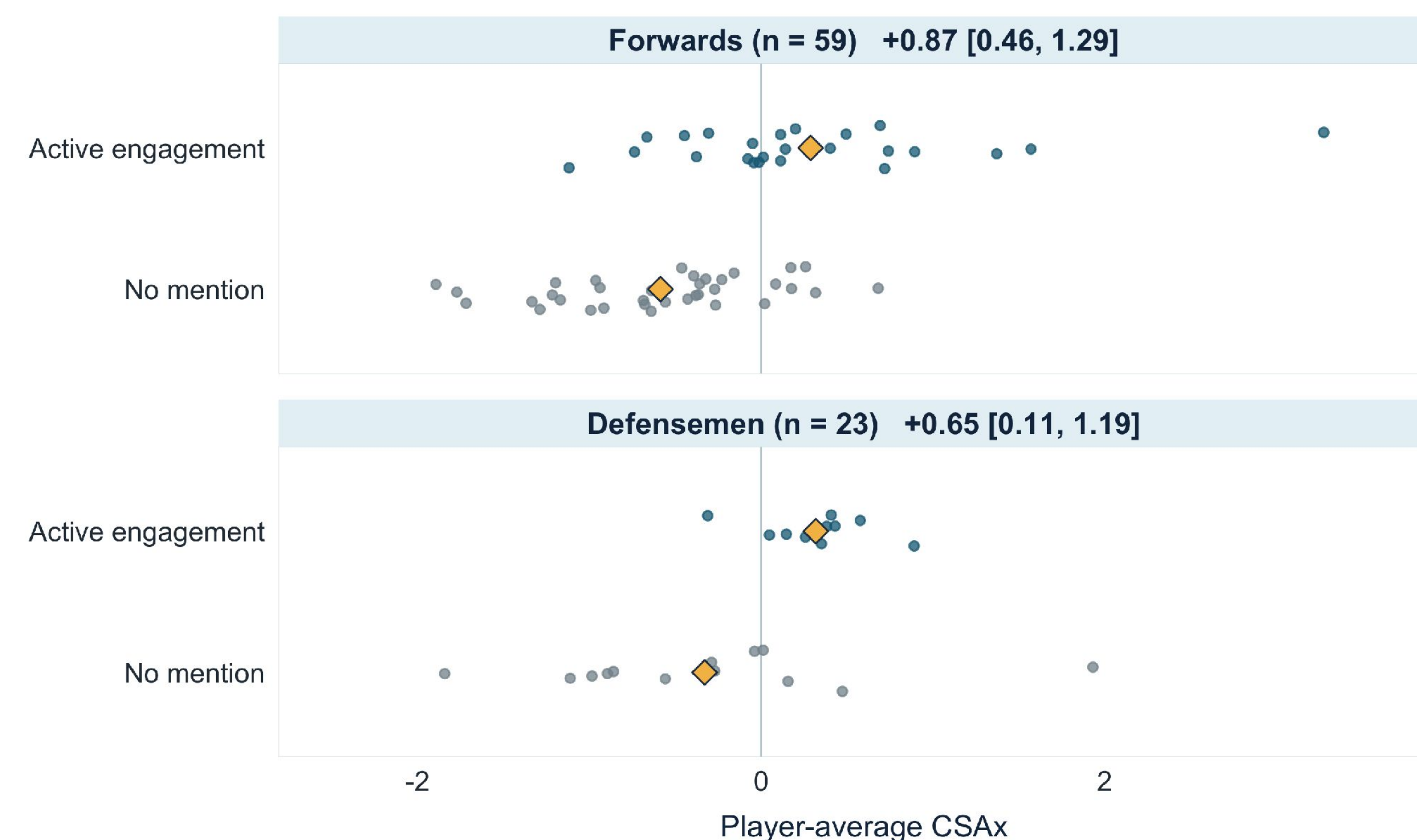
High-CSAx examples (2024–25)

Forwards: Sam Carrick, Mathieu Olivier
Defensemen: Connor Clifton, Radko Gudas

Sources: NHL play-by-play and All Three Zones; 2021–22 to 2024–25.

Does it match what scouts see?

Scouts wrote these descriptions before the NHL seasons we score.

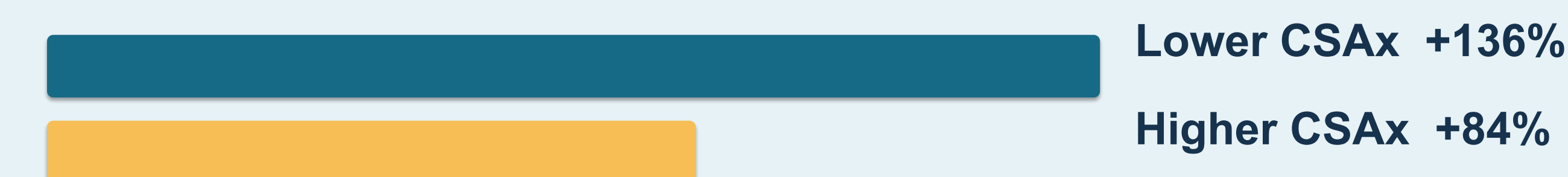


Each dot is one player; diamonds mark group averages. Brackets show 95% intervals for the average difference.

No mention in a report does not mean the scout called the player soft.

What changes in the playoffs?

Forwards: both profiles deliver more hits



Adjusted hit-rate rise, first four playoff games. The defenseman contrast is uncertain.

Where does it matter?

Special teams assign different jobs.

Power play: your team has an extra skater.
Penalty kill: your team plays one skater short.

Higher CSAx is linked to slightly less power-play time and more penalty-kill time in both positions.

Change in share of total ice time

	Power play	Penalty kill
Forwards	–1.2 points	+1.4 points
Defensemen	–0.9 points	+1.3 points

Comparison: CSAx +1 versus –1

More likely to return

Chance of reaching 300 NHL minutes the next season:

**+2.9 points
for forwards**

**+4.2 points
for defensemen**

Among players who return, higher CSAx does not show a clear increase in minutes per game.



2025 NHL intern cohort

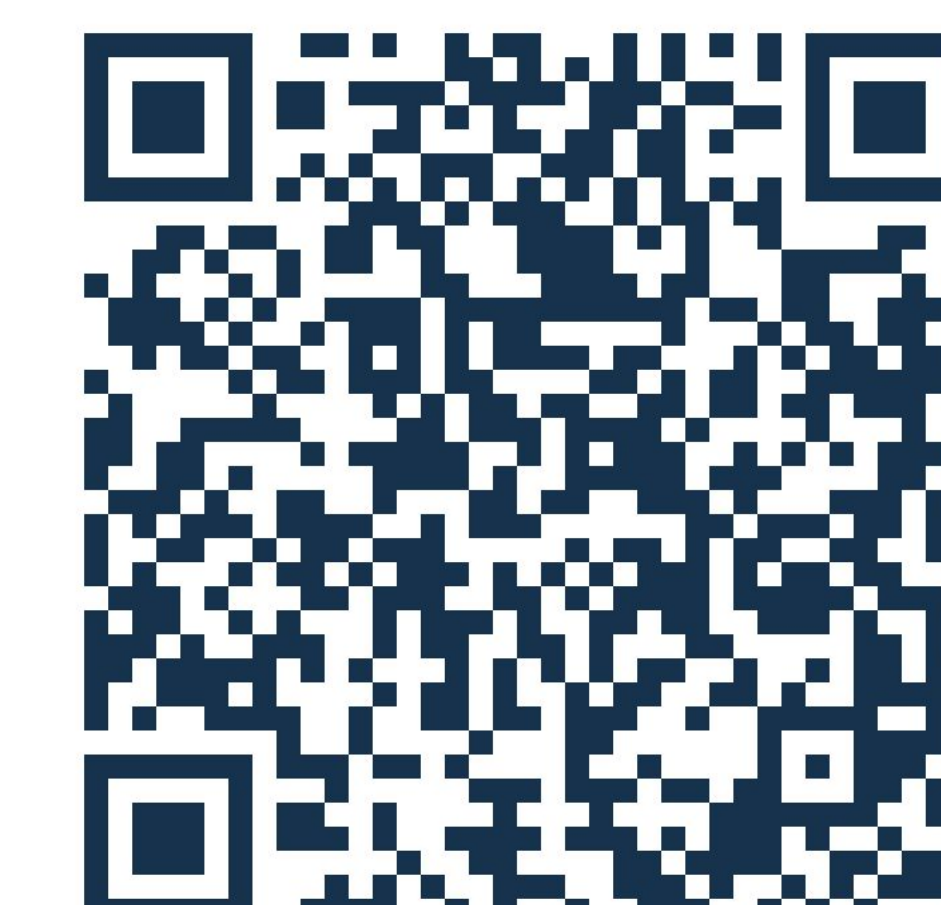
NHL Stats R&D intern cohort

During my NHL internship, conversations with the Stats R&D team helped sharpen the hockey questions behind this study.

Thank you to NHL Stats R&D supervisors and colleagues for their advice and generosity.

Next: Carnegie Mellon poster; MIT Sloan submission; journal paper.

The research continues with new seasons and better observations of puck battles.



**Full 2024–25
rankings and
technical details.**